

Cross-Linguistic Influence of High Rising Terminals: Insights from English to Uyghur

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November 2025

Abstract

High Rising Terminals (HRTs), also known as uptalk, represent a prosodic phenomenon where declarative utterances end with a rising intonation, often perceived as question-like. This paper explores the cross-linguistic influence of HRT, originating prominently in English varieties, on languages such as Uyghur, Spanish, and Serbian. Drawing from referenced studies and web-sourced materials, we provide a mathematical formulation of HRT using fundamental frequency (F0) contours. We discuss its traditional use in Uyghur for interrogatives and its emerging adoption in declarative contexts due to English influence, alongside similar patterns in other languages. Insights highlight sociolinguistic factors like migration and media exposure driving this shift, with implications for language contact theories.

1 Introduction

High Rising Terminals (HRTs) are a distinctive intonational feature in various English dialects, characterized by a pitch rise at the end of non-interrogative utterances [Arvaniti, 2014]. Originally documented in Australian, New Zealand, and Southern Californian English, HRTs serve pragmatic functions such as seeking confirmation or holding the floor in conversation [Warren, 2016]. Referenced studies on HRT in English [Volante, 2023, Ritchart & Arvaniti, 2014, Spindler, 2003] underscore its association with gender, race, and social status, often stigmatized as indicating uncertainty.

In non-English languages, HRT-like patterns traditionally mark questions or imperatives, but recent cross-linguistic influences from English-speaking environments have led to their adoption in declaratives. This is evident in Uyghur, a Turkic language spoken primarily in Xinjiang, China, where HRT is historically reserved for interrogatives but is increasingly observed in declarative speech among diaspora communities in Western countries [Major, 2019]. Similar influences appear in Spanish heritage speakers [Henriksen, 2023] and Serbian

EFL learners [Ilić, 2023]. This paper formulates HRT mathematically, reviews its manifestations in English, and analyzes its spread to other languages, emphasizing sociolinguistic drivers.

2 Definition and Mathematical Formulation of HRT

HRT is phonetically defined as an upward shift in pitch at the utterance boundary, contrasting with the typical falling intonation in declaratives. In the ToBI (Tones and Break Indices) framework, HRT in English is often transcribed as L* H-H% or L* L-H%, where L* denotes a low pitch accent on the stressed syllable, followed by a high boundary tone [Fletcher et al., 2005].

Mathematically, HRT can be modeled using the fundamental frequency (F0) contour over time. Let $F_0(t)$ represent the pitch at time t , where t_0 is the onset of the final stressed vowel, and t_{end} is the utterance end. A simple linear model for the rising contour is:

$$F_0(t) = F_{0,\min} + \alpha(t - t_0), \quad t \in [t_0, t_{\text{end}}],$$

where $F_{0,\min}$ is the minimum F0 at t_0 , and $\alpha > 0$ is the rise rate (in Hz/s). The excursion ΔF_0 , a key measure of HRT prominence, is:

$$\Delta F_0 = F_0(t_{\text{end}}) - F_{0,\min}.$$

Empirical studies report ΔF_0 ranging from 50–150 Hz in English HRT, with alignment (distance from t_0 to rise start) varying by discourse function [Ritchart & Arvaniti, 2014, Levon, 2018]. For instance, in Southern Californian English, statements show smaller ΔF_0 (e.g., 80 Hz) and later alignment compared to questions (120 Hz, earlier onset) [Ritchart & Arvaniti, 2014].

The influence of HRT can be quantified via adoption models. Let $P_l(t)$ be the probability of HRT use in language l at time t , influenced by contact with English. A logistic diffusion model approximates spread:

$$\frac{dP_l}{dt} = rP_l(1 - P_l) + \beta C_E,$$

where r is the intrinsic growth rate, β is the influence coefficient, and C_E is English contact intensity (e.g., via migration or media). This captures gradual increases observed in bilingual contexts [Buck, 2015].

3 HRT in English Varieties

In English, HRT is prevalent in varieties like Southern Californian, Australian, and New Zealand English. The study by Ritchart and Arvaniti [Ritchart & Arvaniti, 2014] analyzes 23 speakers, finding females exhibit greater ΔF_0 and later alignment than males, with no significant ethnic or class effects. Volante’s

thesis [Volante, 2023] links HRT to perceptions of trustworthiness and confidence, influenced by speaker race and gender.

Spindler’s study [Spindler, 2003] explores HRT in American English, noting its role in floor-holding. Cross-dialectally, HRT tunes vary: Australian English uses L* H-H% or H* H-H% [Fletcher et al., 2005], while Glaswegian English employs L*H H-L% [Mayo et al., 1997]. These patterns influence global English, spreading via media and migration [Warren, 2016].

4 Cross-Linguistic Influence of HRT

English HRT exerts influence on other languages through bilingualism and globalization. In Spanish–English heritage bilinguals in Southern California, uptalk transfers to Spanish declaratives, with intonational phrase-final deaccenting and rising boundaries mirroring English patterns [Henriksen, 2023]. Buck [Buck, 2015] documents systematic L1 English to L2 Spanish transfer, increasing HRT frequency in bilingual speech.

In Serbian EFL learners, uptalk in L2 English influences native Serbian intonation, with phonetic realization showing hybrid F0 contours [Ilić, 2023]. Cross-linguistic studies suggest English media (e.g., TV shows) accelerates adoption [Warren, 2016].

For Uyghur, a stress-accent language with edge-marking intonation [Major, 2019], HRT traditionally marks interrogatives or imperatives, using duration and intensity over F0 for stress [Engesæth et al., 2019]. However, among Uyghur diaspora in English-speaking countries, declarative HRT emerges due to code-switching and acculturation, paralleling Spanish patterns. Though direct empirical studies on Uyghur are limited, analogous influences in related Turkic languages (e.g., Kazakh) via English contact support this trend [Warren, 2016]. Migration to the West increases C_E , elevating P_l per the diffusion model.

This influence underscores language contact dynamics, where prosodic features spread faster than lexical ones due to subconscious mimicry [Levon, 2018].

5 Conclusion

HRT exemplifies how English prosody influences global languages, from mathematical F0 models to sociolinguistic adoption. In Uyghur and other languages, this shift from interrogative to declarative use highlights globalization’s role. Future research should empirically quantify ΔF_0 in Uyghur bilinguals to validate and refine the proposed models.

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